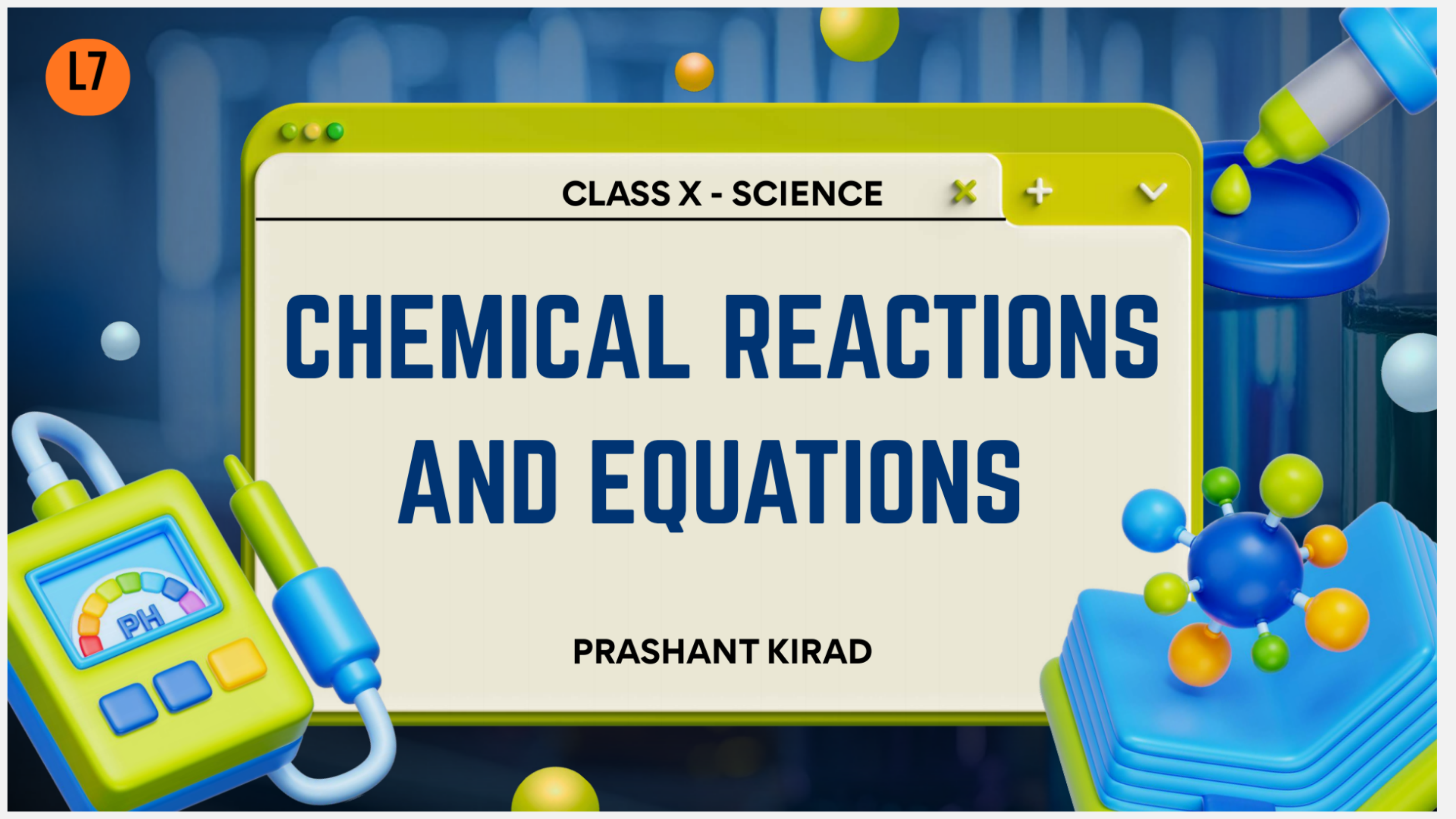


L7

CLASS X - SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

PRASHANT KIRAD



LET'S TEST YOUR MEMORY



Q. The nitrate compound of a metal 'X' undergoes a precipitation reaction to produce a white precipitate of the chloride of the same metal 'X'. This metal in its nitrate form also reacts with potassium iodide (KI) to form a yellow precipitate, with potassium nitrate as a by-product.

On the basis of this information, answer the following:

(A) Identify the metal 'X' and its nitrate compound.

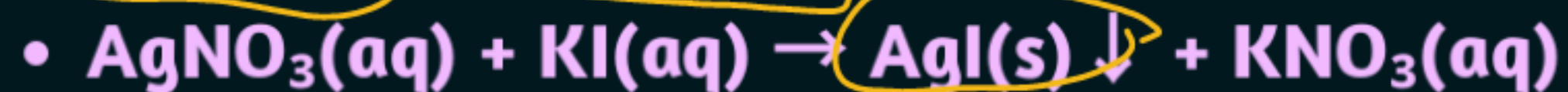
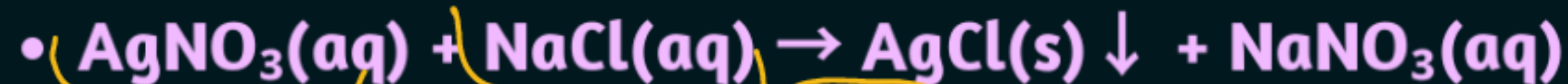
(B) Write balanced chemical equations with correct states of matter for both reactions.

Ans -

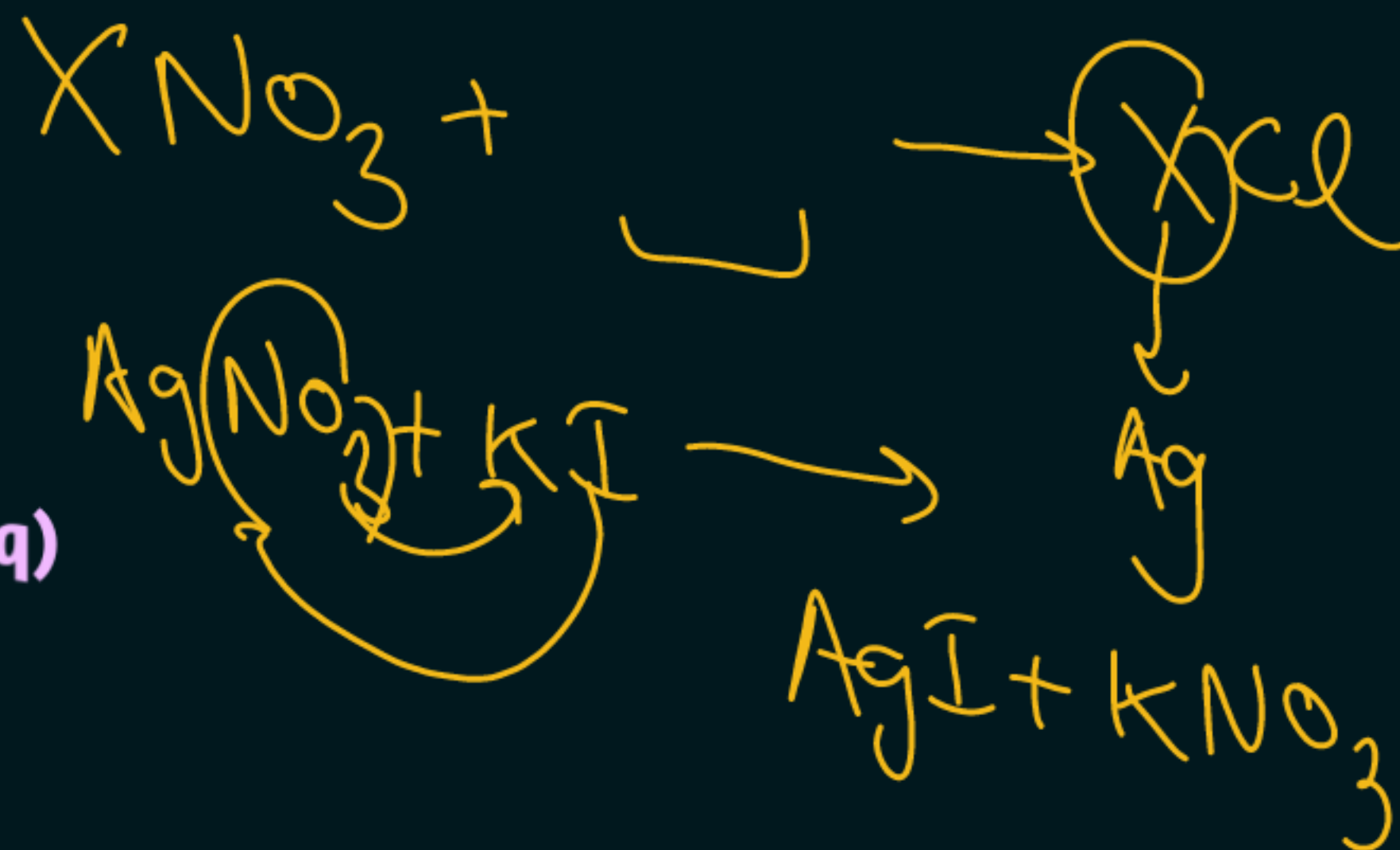
(A) Metal 'X' is Silver (Ag).

Its nitrate compound is Silver nitrate (AgNO_3).

(B)



(White precipitate: AgCl , Yellow precipitate: AgI)



LET'S TEST YOUR MEMORY



Q. When X reacts with Y, a yellow precipitate of Z is formed along with a colourless solution of potassium nitrate. Identify X, Y, Z and the type of reaction.

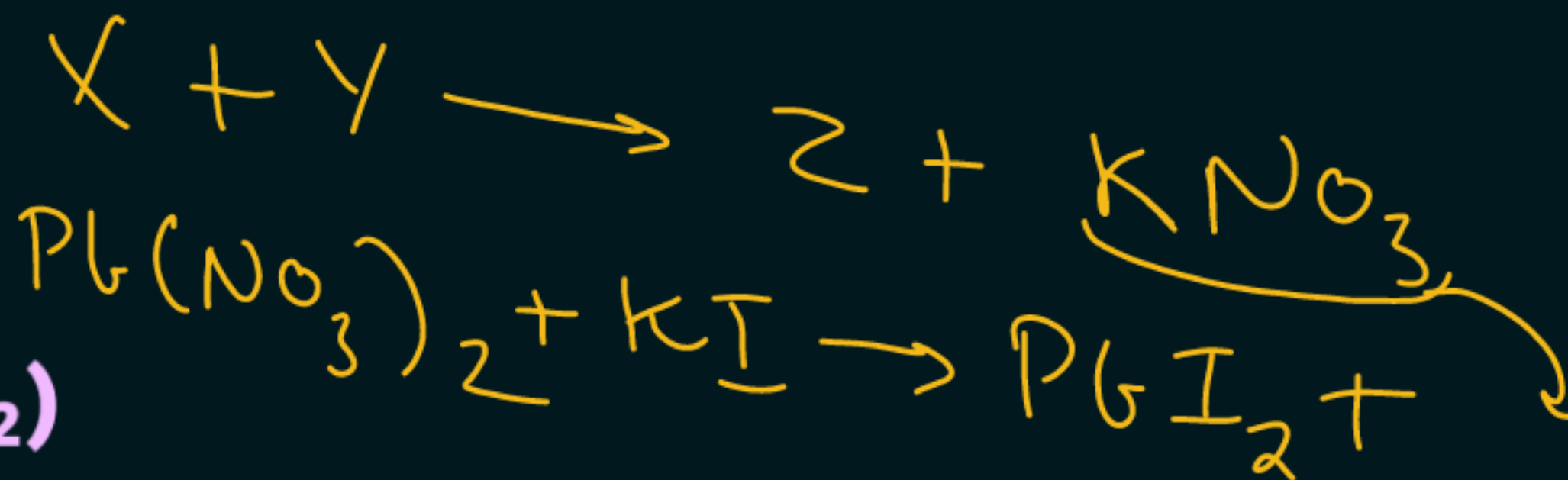
Ans -

X = Lead nitrate ($\text{Pb}(\text{NO}_3)_2$)

Y = Potassium iodide (KI)

Z = Lead iodide (PbI_2) – yellow precipitate

Type of reaction: Double displacement reaction (precipitation reaction)



LET'S TEST YOUR MEMORY



Q. An aqueous solution of metal nitrate P reacts with sodium bromide solution to form yellow precipitate O used in photography. On exposure to sunlight, O decomposes to form metal P along with reddish-brown gas. Identify P & O.

Write the reaction and type of reaction.

Ans -

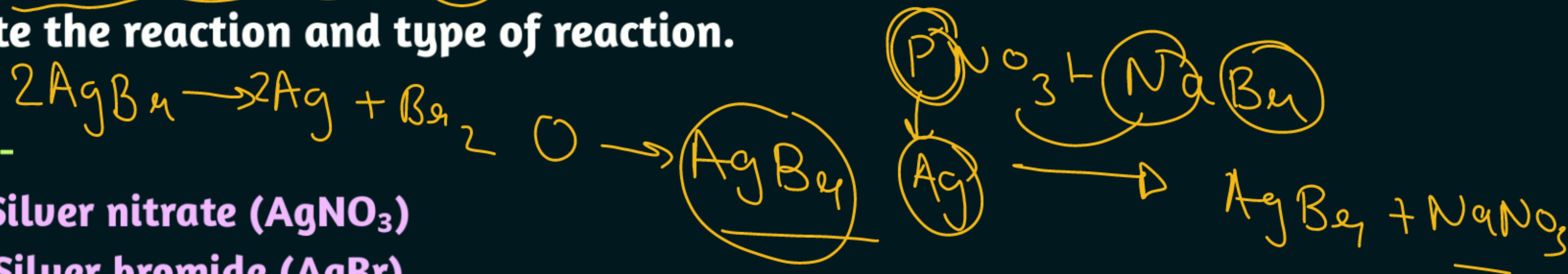
P = Silver nitrate (AgNO_3)

O = Silver bromide (AgBr)

Reaction 1 (Double Displacement): $\text{AgNO}_3(\text{aq}) + \text{NaBr}(\text{aq}) \rightarrow \text{AgBr}(\text{s}) + \text{NaNO}_3(\text{aq})$

Reaction 2 (Photochemical Decomposition): $2\text{AgBr}(\text{s}) \rightarrow 2\text{Ag}(\text{s}) + \text{Br}_2(\text{g})$ (sunlight)

Types of reaction: Double displacement and Photochemical decomposition (Redox)



LET'S TEST YOUR MEMORY



Q. Rehan takes two test tubes. In the first test tube, he takes lead nitrate solution and in the second, he takes a few zinc granules. He adds potassium iodide solution to the lead nitrate solution in the first test tube. He adds dilute sulphuric acid in the second test tube containing zinc granules.

(A) What is observed in the first test tube on adding potassium iodide solution?

(B) When dilute sulphuric acid is added to a few zinc granules in the second test tube:

☒ (I) Bubbles of hydrogen gas can be seen.

(II) Bubbles of carbon dioxide can be seen around zinc granules.

☒ (III) The flask becomes hot.

(IV) Colour of solution changes to blue.

Select the correct observations:

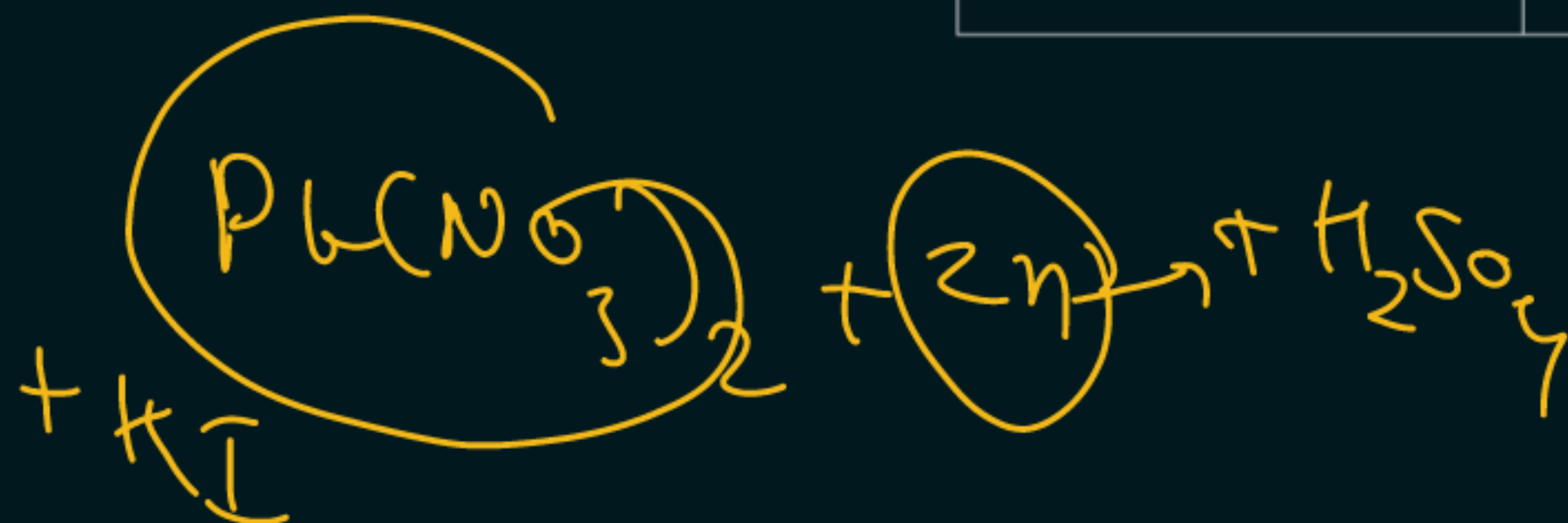
(a) Only (I)

☒ (b) Both (I) and (III)

(c) Both (II) and (III)

(d) (I), (III) and (IV)

Formula of Precipitate	Colour of Precipitate
(a) KNO_3	Yellow
(b) KNO_3	Brown
(c) PbI_2	Yellow
(d) PbI_2	Brown



(C) Assertion (A): A single displacement reaction takes place when zinc granules are added to dilute sulphuric acid.

Reason (R): Hydrogen gas is evolved in the reaction and there is no change in colour. $\uparrow$

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

~~(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).~~

(c) (A) is true but (R) is false.

(d) (A) is false but (R) is true.



(D) Suppose Rehan added dilute hydrochloric acid in place of sulphuric acid in the second test tube. Select the correct observation made by him:

(a) Chlorine gas will be evolved.

(b) The test tube will become cold as heat is absorbed during this reaction.

~~(c) Hydrogen gas will be evolved.~~

(d) A precipitate of zinc chloride will be formed.



OR

The reaction between lead nitrate and potassium iodide is a double displacement reaction. If a student heats only lead nitrate powder and gets a brown coloured gas, answer the following:

Type of reaction	Name of gas
(a) Decomposition	<u>NO₂</u>
(b) Combination	NO ₂
(c) Decomposition	O ₂
(d) Combination	O ₂



Introduction to Redox Reactions

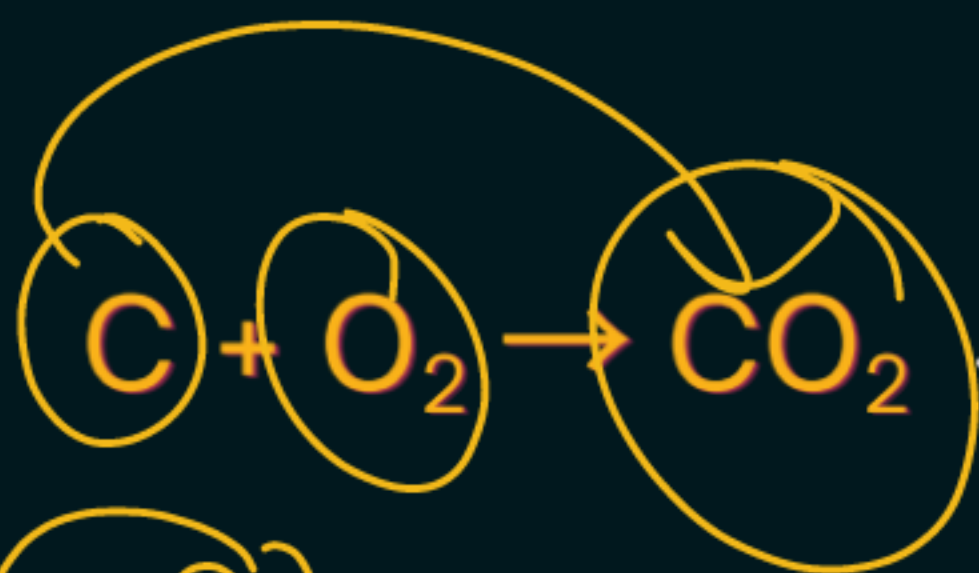


Oxidation

REDOX

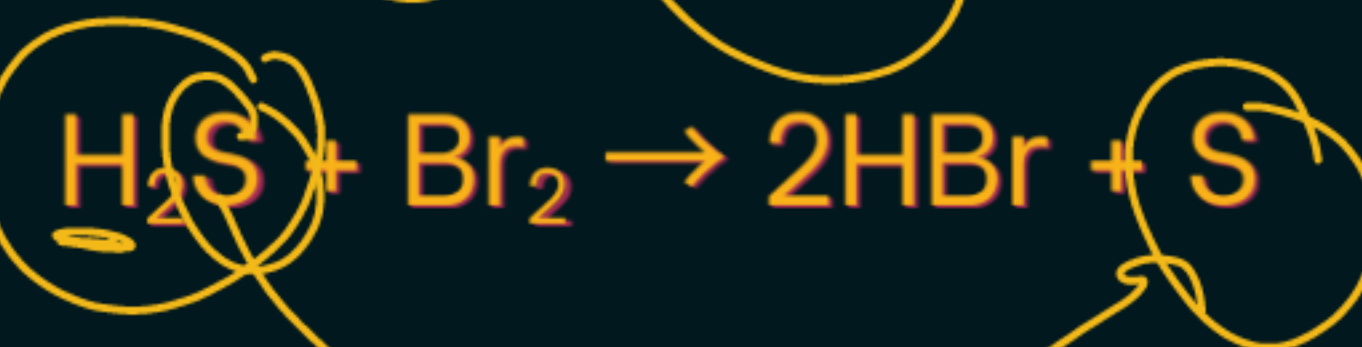
Oxidation refers to the addition of oxygen or the removal of hydrogen from a substance.

Addition of Oxygen:



Oxidation of carbon

Removal of hydrogen:

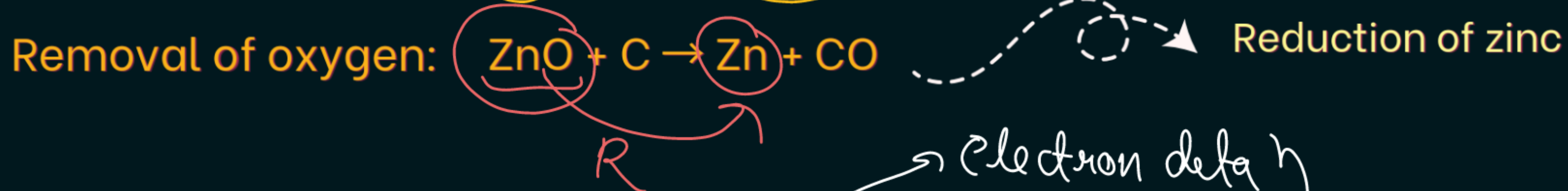
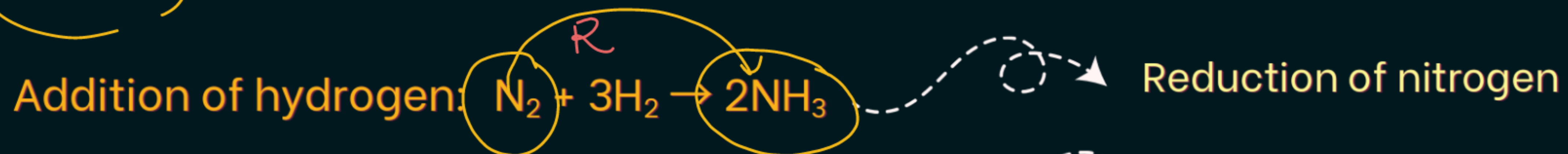


Oxidation of sulphide

Oxidising agent - is a substance that causes oxidation by accepting electrons; therefore, it gets reduced.

Reduction

A chemical reaction which involves addition of hydrogen or removal of oxygen is called as reduction



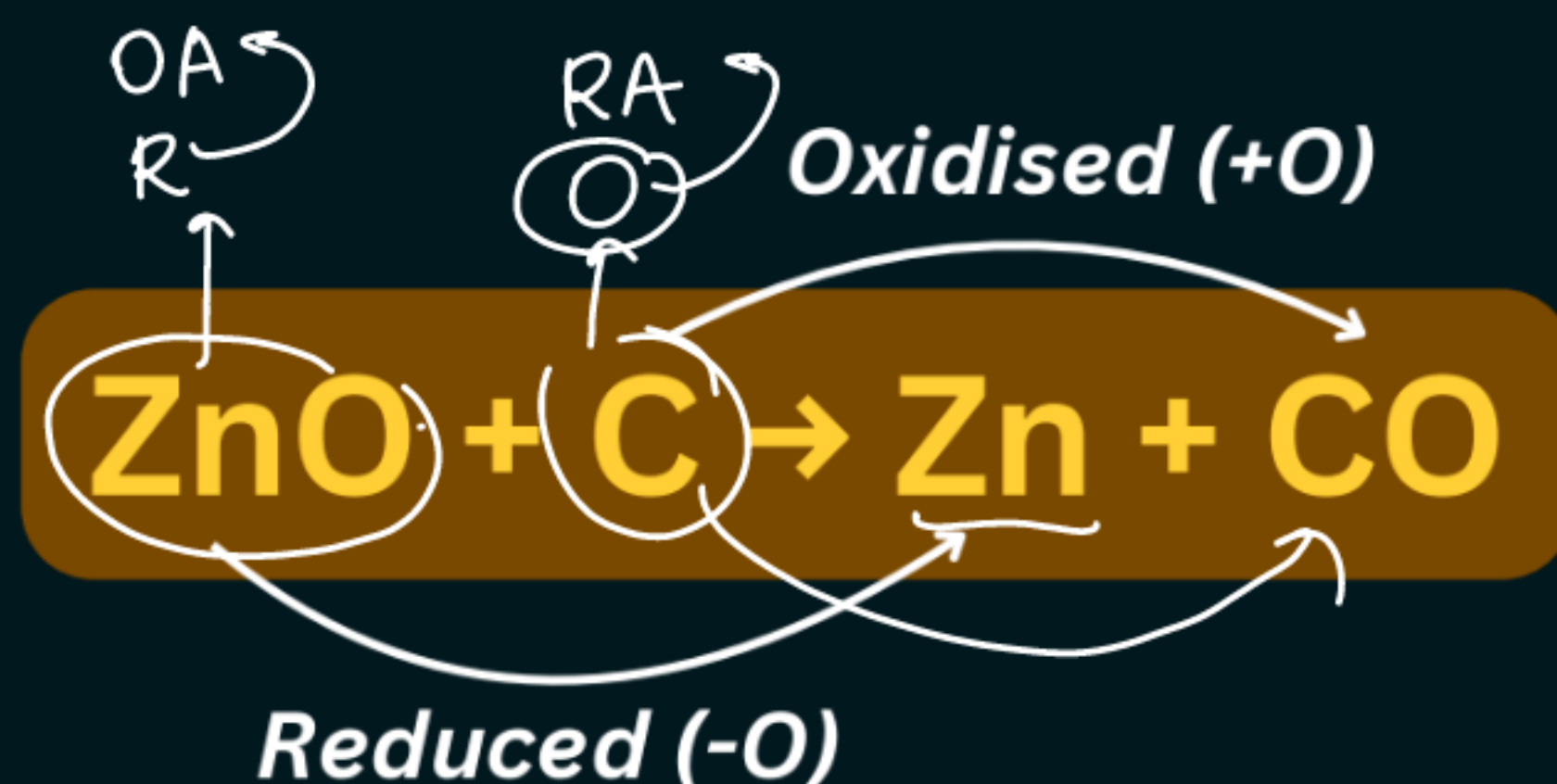
Reducing agent is a substance which brings about reduction. In the above examples H_2 and C are reducing agents.

Electron donor

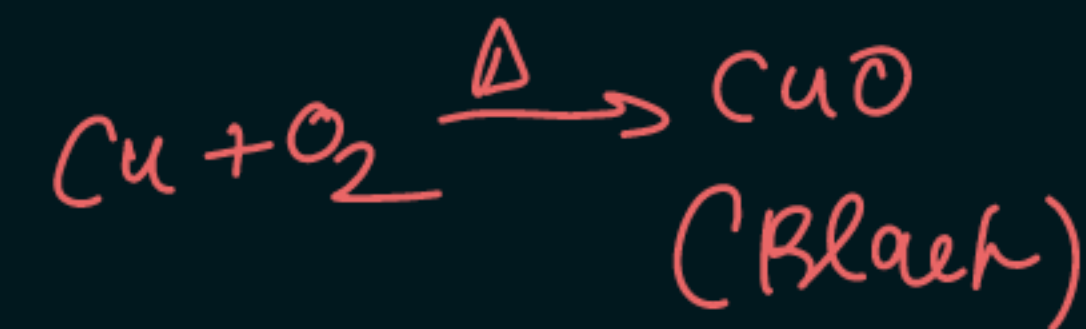
Redox Reaction

A chemical reaction in which reduction and oxidation takes place simultaneously.

In a redox reaction substance getting oxidised is called a Reducing agent. And the substance getting reduced is called an Oxidising agent.



NCERT ACTIVITY



Activity 1.11

- Heat a china dish containing about 1 g copper powder (Fig. 1.10).
- What do you observe?

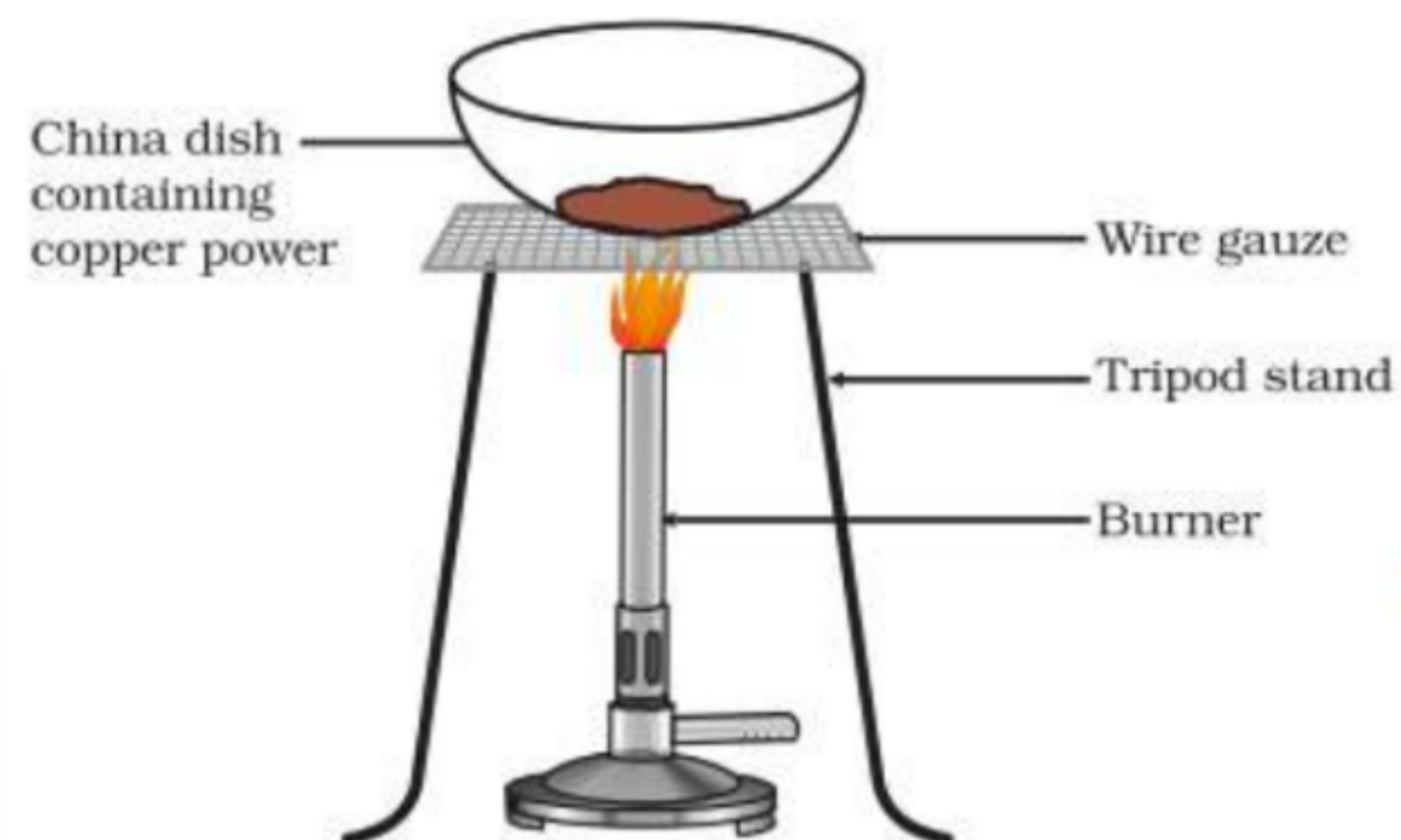


Figure 1.10
Oxidation of copper to copper oxide

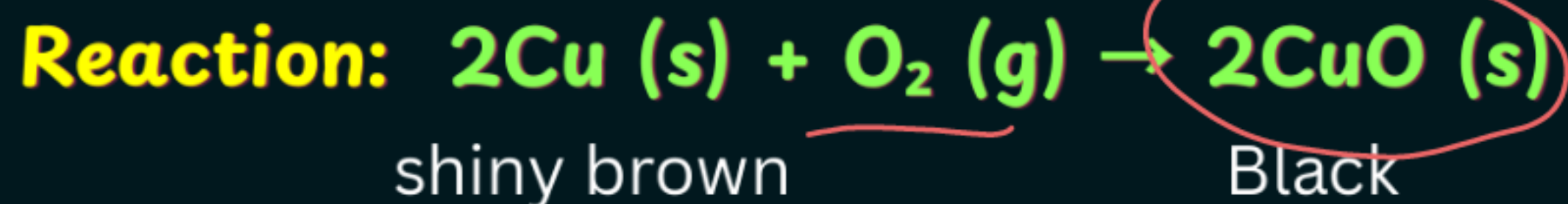
PK Points!



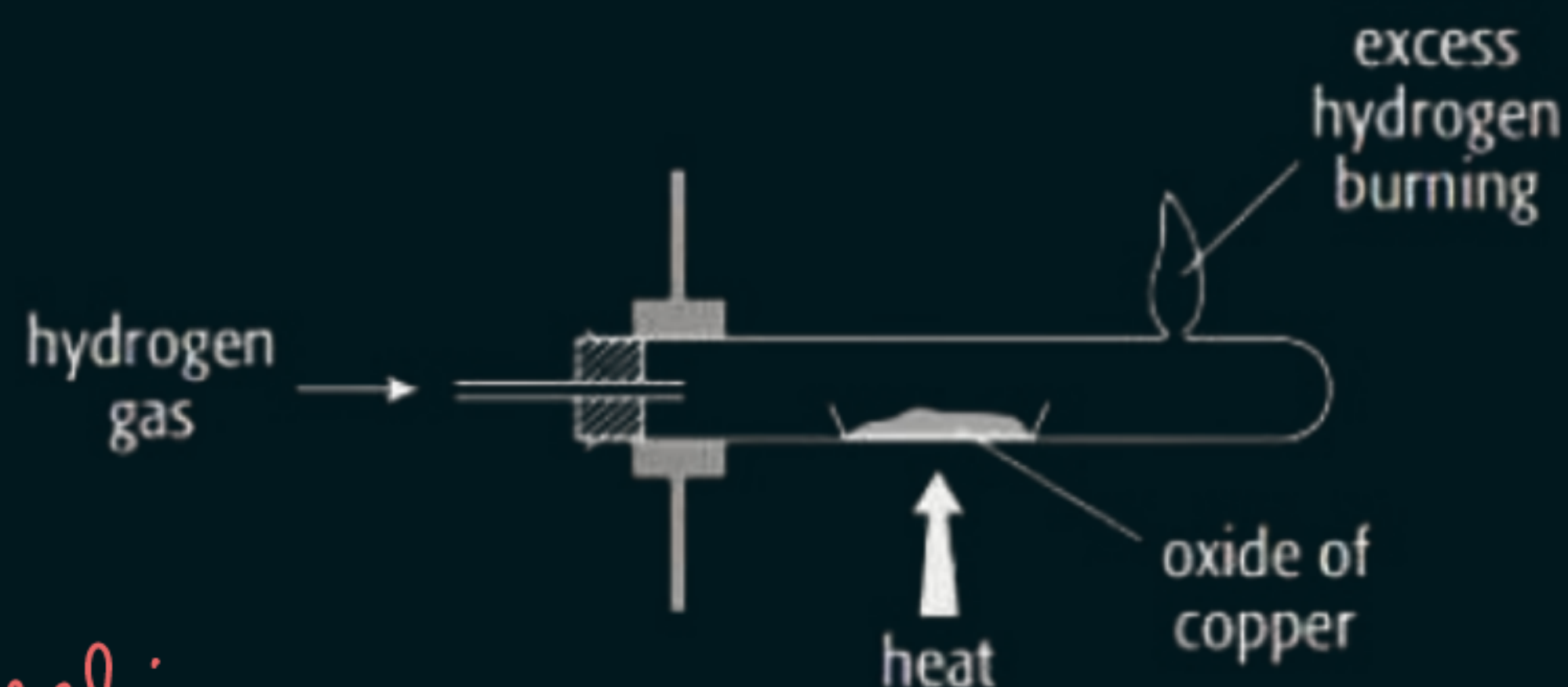
Observation:

- The reddish-brown copper powder slowly turns black.
- This happens because copper reacts with oxygen in the air to form copper oxide.

Conclusion: This is an oxidation reaction. Copper gets oxidized to copper(II) oxide (CuO) when heated in air.



Reduction of copper oxide to copper



Reduction



Redox



LET'S TEST YOUR MEMORY



Q. Which of the following is a redox reaction?
(CBSE)

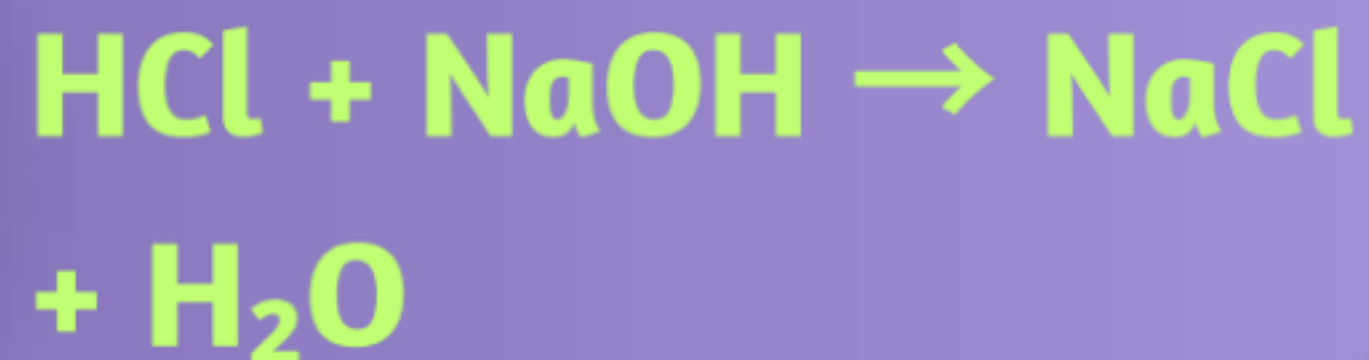
A



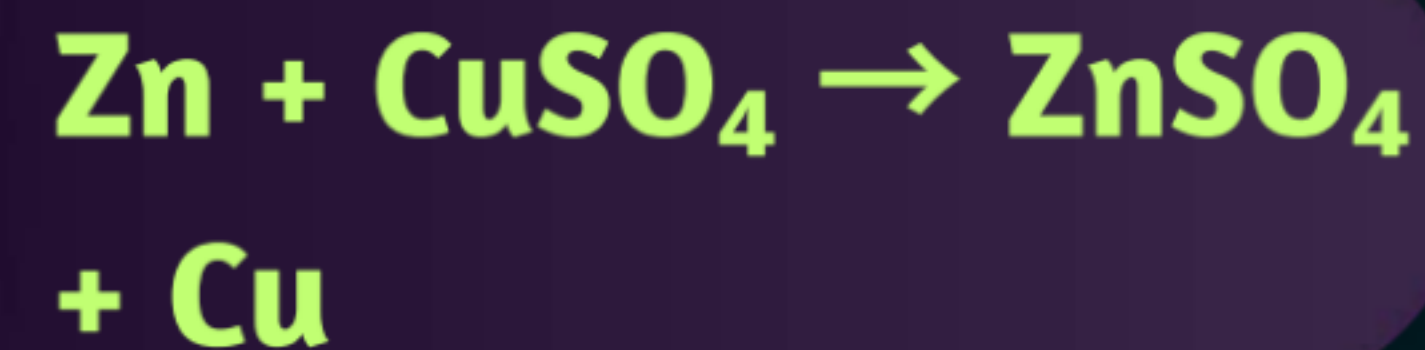
B



C



D



LET'S TEST YOUR MEMORY



Q. Which substance is reduced in the following reaction?



A

CuO

B

H₂

C

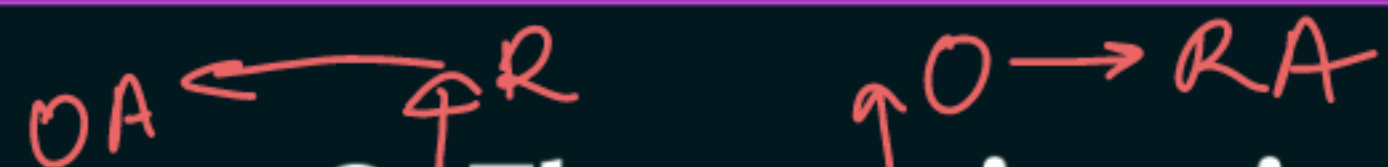
Cu

D

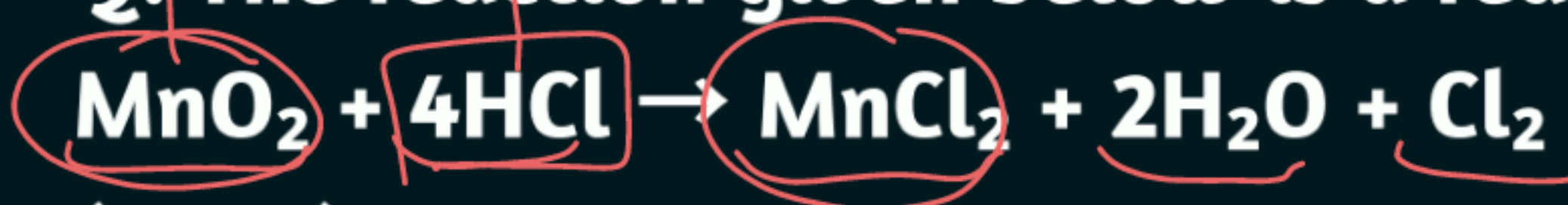
H₂O



LET'S TEST YOUR MEMORY



Q. The reaction given below is a redox reaction because:



(2024)

A

MnO_2 is oxidised and
 HCl is reduced.

B

HCl is oxidised.

C

MnO_2 is reduced.

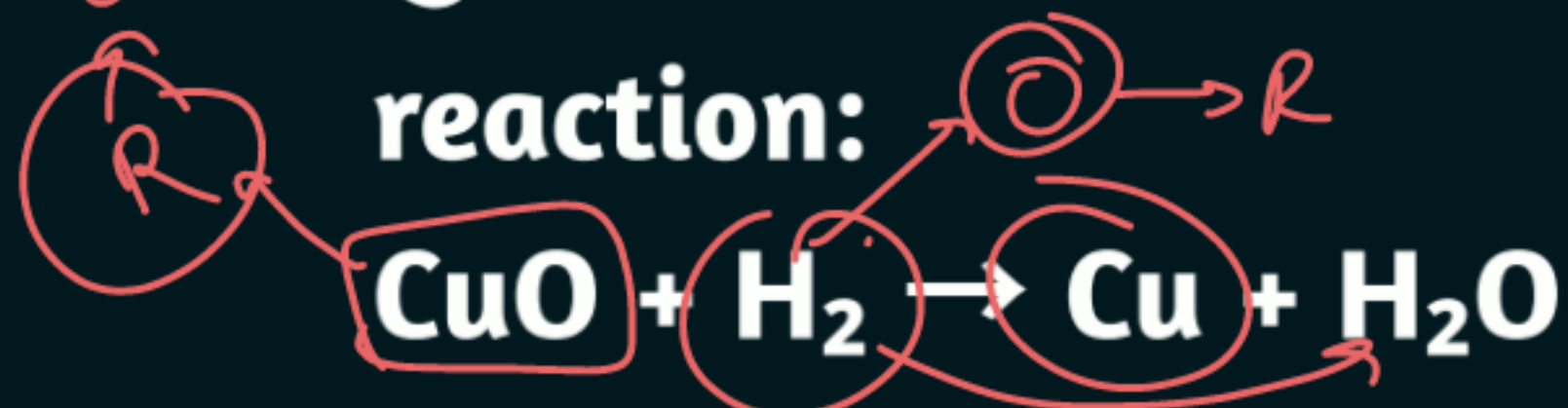
D

MnO_2 is reduced and
 HCl is oxidised.

LET'S TEST YOUR MEMORY



Q. Select the correct statement from the following about the reaction:



- (a) CuO is getting oxidised and H₂ is getting reduced.
- ☒ (b) H₂ is getting oxidised and CuO is getting reduced.
- (c) CuO is a reducing agent.
- (d) H₂ is an oxidising agent.

LET'S TEST YOUR MEMORY



OA ← R → RA

Q. Which of the statements about the reaction below are incorrect?



~~i) Lead is getting reduced.~~

~~ii) Carbon dioxide is getting oxidised.~~

~~iii) Carbon is getting oxidised.~~

~~iv) Lead oxide is getting reduced. [NCERT]~~

~~a) (i) and (ii)~~

b) (i) and (iii)

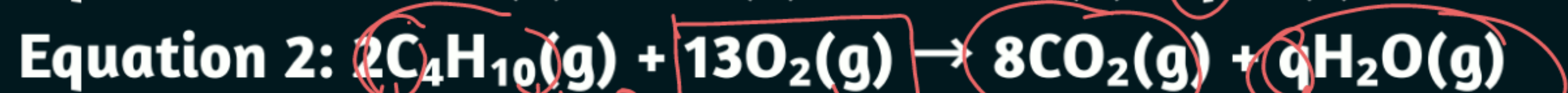
c) (i), (ii) and (iii)

d) All

LET'S TEST YOUR MEMORY



Q. Which of the following equations represent redox reactions and what are the values of 'p' and 'q'?



- (a) Only equation 1 is redox
(b) Both are redox, $p = 2$, $q = 4$
(c) Only equation 2 is redox
(d) Both are redox, $p = 2$, $q = 10$

LET'S TEST YOUR MEMORY



OA ← R O → RA

Q. Identify which substance is oxidised and which is reduced in the reaction: $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Fe}$

Also identify:

- Oxidising agent Fe_2O_3
- Reducing agent Al

LET'S TEST YOUR MEMORY



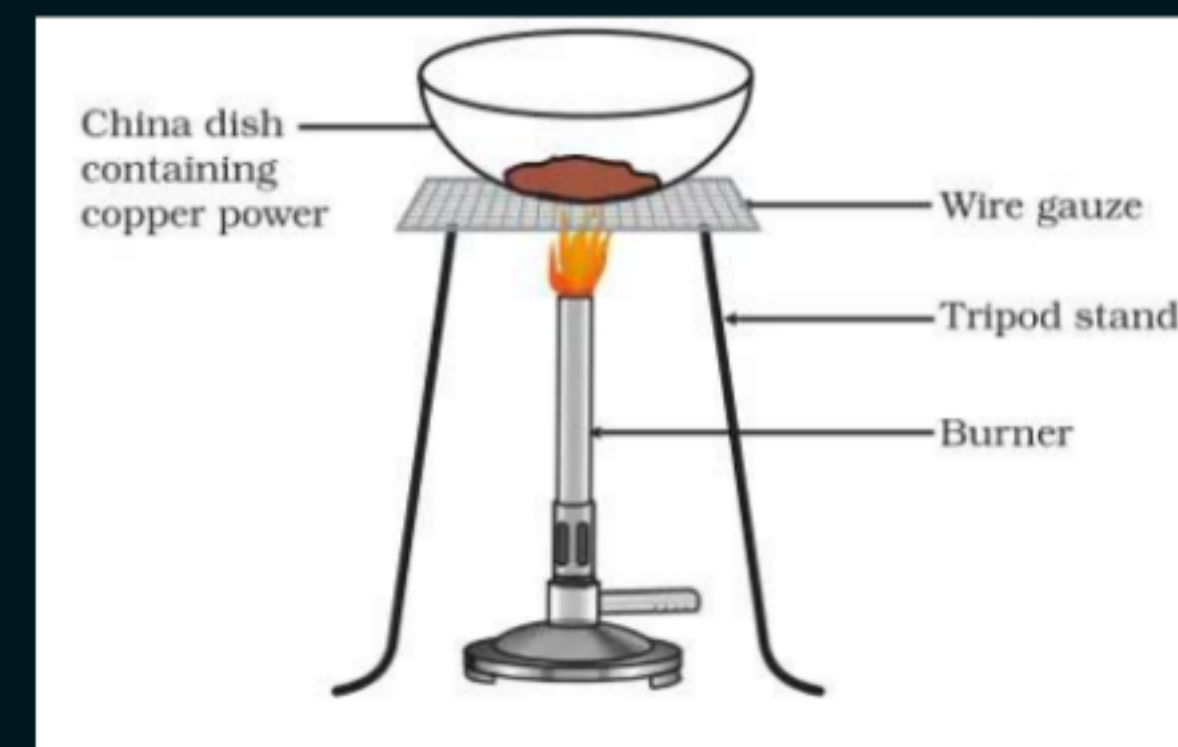
Q. A reddish-brown metal 'Y' reacts with oxygen upon heating to form a black solid 'X'. When 'X' is heated with hydrogen gas, it gets reduced back to the reddish-brown metal 'Y'. Identify X and Y, and write the balanced chemical equations for the reactions.



LET'S TEST YOUR MEMORY



Q. In a chemical experiment, a student heated the copper powder in the china dish by placing it on the Bunsen burner with the help of a wire gauze and a tripod stand, as shown. After some time, it was observed that the copper powder had changed its colour. On the basis of this, answer the following questions:



(A) What has actually happened in the experiment when copper powder was heated? $2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$ (Black)

(B) What if hydrogen gas is passed onto the product formed in (A)? Write your observation.



Ans -

(A) When copper powder is heated in air, it reacts with oxygen and forms black copper(II) oxide (CuO).

Reaction: $2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$

(Colour change: Reddish-brown copper $\rightarrow$ Black copper oxide)

(B) If hydrogen gas is passed over the black copper oxide (CuO), it gets reduced back to copper.

Reaction: $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$

Observation: Black colour changes back to reddish-brown copper, and water droplets are formed.

Home Work



Q. Rahul was asked to perform an experiment by his science teacher. The teacher told Rahul to take some copper powder in a china dish and heat it by placing it on a Bunsen burner with the help of a wire gauze and tripod stand. After some time, Rahul observed that the copper powder had changed its colour.

(A) Observe the given figure and write what happened when Rahul heated the copper powder in a china dish. Explain his observation.

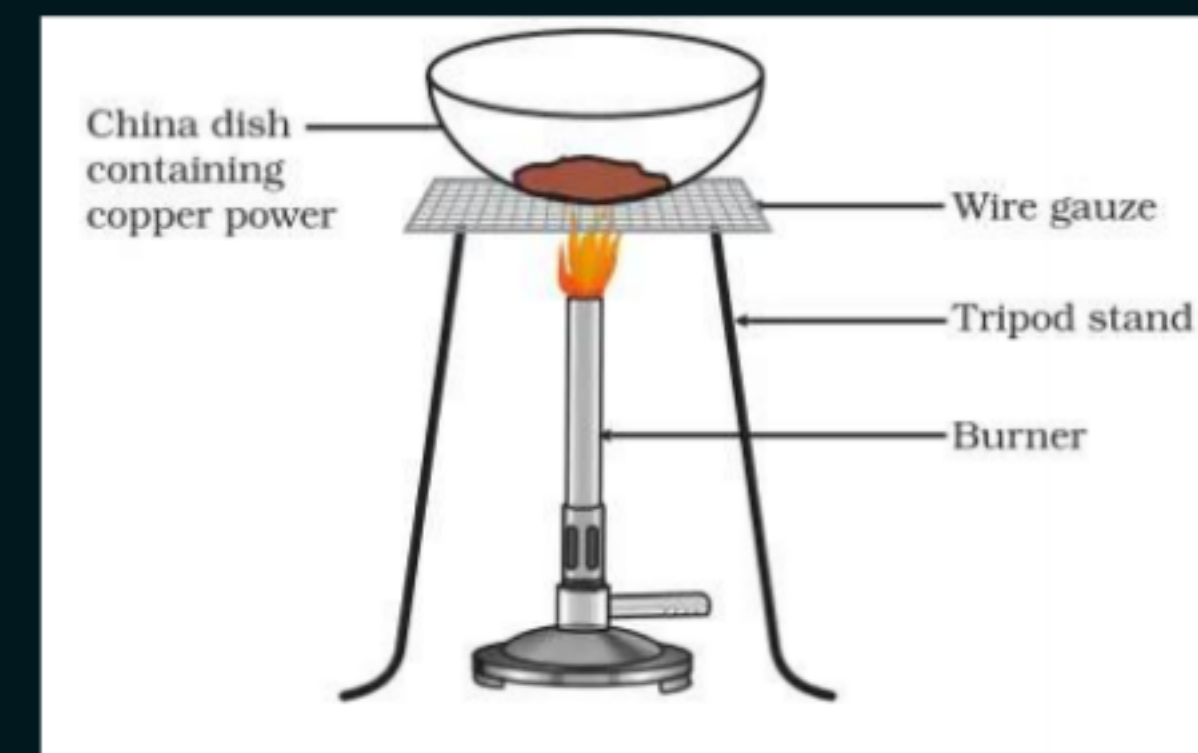
(B) What will you observe if hydrogen gas is passed over the heated material obtained in (A) above?

(C) (i) Identify the substance oxidised and substance reduced in the reaction at (B) above.

(ii) Given is the reaction between lead sulphide and hydrogen peroxide:



Identify the oxidised and reduced compounds.



HW

Characteristics of Chemical Reaction:

Indicators which tell us whether the reaction took place or not:

1. Change in Colour: Example: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$

Blue green

2. Change in Temperature:

a. **Endothermic Reaction (Heat absorbed)**: Example: $\text{CaCO}_3 + \text{Heat} \rightarrow \text{CaO} + \text{CO}_2$

b. **Exothermic Reaction (Heat released)**: Example: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{Heat}$

3. Change in Stats: Example: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ (l)

4. Evolution of Gas: Example: $\text{Zn}^{(g)} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$

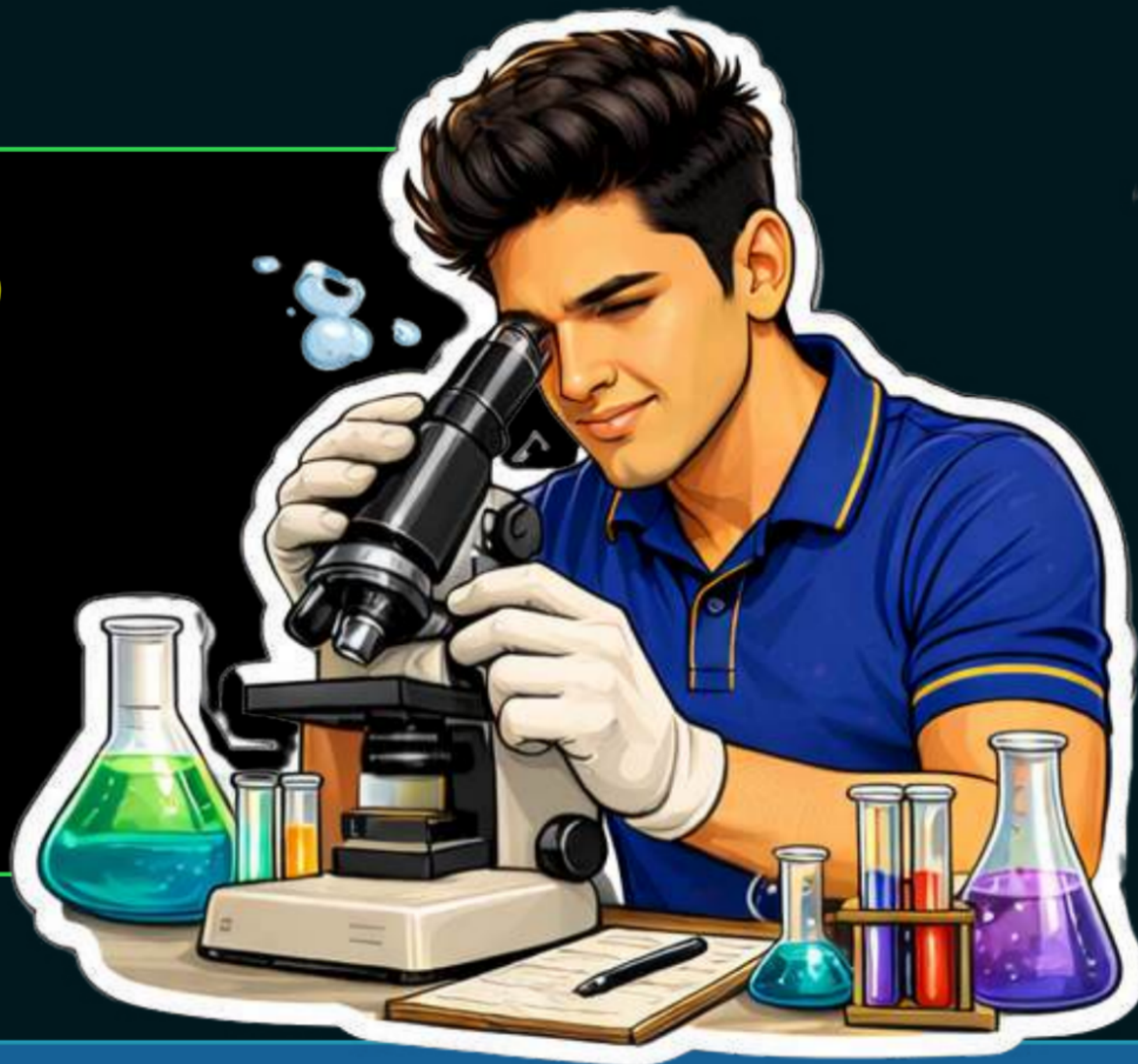
✓ 5. Formation of Precipitate: Example: $\text{Pb(NO}_3)_2 + \text{KI} \rightarrow \text{PbI}_2 + \text{KNO}_3$



RECAP



Reaction Rapid Revision



Chemical Equation	Type of Reaction	Important Key Point
$\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$	Combination	Bright white dazzling light
$\text{CaO(s)} + \text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)} + \text{Heat}$	Combination, Exothermic	Heat released, forms slaked lime
$\text{Ca(OH)}_2\text{(aq)} + \text{CO}_2\text{(g)} \rightarrow \text{CaCO}_3\text{(s)} + \text{H}_2\text{O(l)}$	Combination	Milky white precipitate (limewater test)
$2\text{FeSO}_4\text{(s)} \rightarrow \text{Fe}_2\text{O}_3\text{(s)} + \text{SO}_2\text{(g)} + \text{SO}_3\text{(g)} \text{ (Heat)}$	Decomposition	Smell of SO_2 gas (choking smell), residue is brown
$\text{CaCO}_3\text{(s)} \rightarrow \text{CaO(s)} + \text{CO}_2\text{(g)} \text{ (Heat)}$	Decomposition	Gas evolved (CO_2), residue is quick lime
$2\text{Pb(NO}_3)_2\text{(s)} \rightarrow 2\text{PbO(s)} + 4\text{NO}_2\text{(g)} + \text{O}_2\text{(g)} \text{ (Heat)}$	Decomposition	Brown fumes of NO_2 gas
$\text{H}_2\text{O} \rightarrow \text{H}_2 + \text{O}_2 \text{ (Electricity)}$	Decomposition	Volume of H_2 gas collected is double the volume of O_2 gas.
$2\text{AgCl(s)} \rightarrow 2\text{Ag(s)} + \text{Cl}_2\text{(g)} \text{ (Sunlight)}$	Photochemical Decomposition	White AgCl turns grey

Chemical Equation	Type of Reaction	Important Key Point
✓ $2\text{AgBr(s)} \rightarrow 2\text{Ag(s)} + \text{Br}_2\text{(g)}$ (Sunlight)	Photochemical Decomposition	Pale yellow AgBr turns grey
✓ $\text{Fe(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{FeSO}_4\text{(aq)} + \text{Cu(s)}$	Displacement	Blue color of CuSO_4 fades, reddish brown copper deposited
✓ $\text{Na}_2\text{SO}_4\text{(aq)} + \text{BaCl}_2\text{(aq)} \rightarrow \text{BaSO}_4\text{(s)} + 2\text{NaCl(aq)}$	Double Displacement	White precipitate of BaSO_4 formed
✓ $\text{Pb(NO}_3)_2\text{(aq)} + \text{KI(aq)} \rightarrow \text{PbI}_2\text{(s)} + \text{KNO}_3\text{(aq)}$	Double Displacement	Yellow precipitate of PbI_2 formed
✓ $2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$ (Heat)	Combination	Black coating of CuO forms
$\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$	Redox	Black CuO reduced to reddish brown copper